

MA1002 — Calculus for Science Part 2

Summer 2025 Examination · Paper 1 · Full Marking Scheme

Module	MA1002 — Calculus for Science Part 2
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Institution	University College Cork
Session	Summer 2025, Paper 1
Duration	1.5 hours (+ 15 minutes reading time)
Structure	Answer any 4 of the 5 questions . All questions carry equal marks.
Materials	Formula sheet supplied (calculus, trigonometry, length/area/volume).

How to use this scheme. Every answer is worked out in small numbered steps, with a “Rules you need” box listing the rules used. Nothing is skipped, so you can see where every number comes from. Read the answer first, then follow the steps.

Coverage note. On the day you answer 4 questions, but this scheme works through **all five** in full, so you can pick the ones you’re strongest on.

Source basis. Every answer uses only MA1002 course methods: indefinite integrals, substitution, integration by parts, definite integrals and area, volumes of rotation, partial fractions and separable differential equations. Q5(b) (line of best fit) isn’t a listed course topic; the question supplies its formulas, and a note marks it.

Mark splits. The paper doesn’t print marks for each part. This scheme gives each question 25 marks. The split between parts and within each part is Study-Uni’s suggested allocation.

Own-figure rule. Where a later step uses an earlier result, a correct method applied to a wrong earlier value still earns the method marks.

Rounding. Answers are given exactly (e.g. $32/3$, $3\pi/10$, $-11/384$), with a decimal to 3 significant figures where useful. Q5(b) is rounded to 1 decimal place as the question asks.

Question 1

Antiderivatives (the indefinite integral), initial conditions — 25 marks

1(a)

[13 marks]

Find all functions $f(x)$ such that $f'(x) = 4 \sin(2x) - e^{2x} + (\sqrt{x} + \frac{1}{\sqrt{x}})^2$.

ANSWER

$$f(x) = -2 \cos(2x) - \frac{1}{2}e^{2x} + \frac{x^2}{2} + 2x + \ln|x| + C$$

RULES YOU NEED

- $\int \sin(ax) \, dx = -(1/a) \cos(ax)$
- $\int e^{ax} \, dx = (1/a) e^{ax}$
- $\int x^n \, dx = x^{n+1}/(n+1)$ ($n \neq -1$), and $\int (1/x) \, dx = \ln|x|$
- "All functions" means add $+ C$: any constant differentiates to 0.

WORKING — STEP BY STEP

1 Expand the bracket first

There is no rule for integrating a bracket squared, so multiply it out using $(a + b)^2 = a^2 + 2ab + b^2$, with $a = \sqrt{x}$ and $b = 1/\sqrt{x}$:

$$a^2 = (\sqrt{x})^2 = x \quad 2ab = 2 \cdot \sqrt{x} \cdot \frac{1}{\sqrt{x}} = 2 \quad b^2 = \left(\frac{1}{\sqrt{x}}\right)^2 = \frac{1}{x}$$

$$(\sqrt{x} + \frac{1}{\sqrt{x}})^2 = x + 2 + \frac{1}{x}$$

2 Rewrite $f'(x)$ as simple terms

$$f'(x) = 4 \sin(2x) - e^{2x} + x + 2 + \frac{1}{x}$$

3 Integrate term by term

Take each term on its own. Here $a = 2$ in both $\sin(2x)$ and e^{2x} :

$$4 \sin(2x) \rightarrow 4 \times \left(-\frac{1}{2} \cos(2x)\right) = -2 \cos(2x)$$

$$-e^{2x} \rightarrow -\frac{1}{2} e^{2x}$$

$$x \rightarrow \frac{x^2}{2} \quad 2 \rightarrow 2x \quad \frac{1}{x} \rightarrow \ln|x|$$

4 Add the constant

The question asks for **all** functions, so add + C:

$$f(x) = -2 \cos(2x) - \frac{1}{2}e^{2x} + \frac{x^2}{2} + 2x + \ln|x| + C$$

(Since $\sqrt{x}$ only makes sense for $x > 0$, $\ln|x|$ can also be written $\ln x$.)

5 Check (optional)

Differentiate back: $-2 \cos(2x) \rightarrow 4 \sin(2x)$, $-\frac{1}{2}e^{2x} \rightarrow -e^{2x}$, $x^2/2 \rightarrow x$, $2x \rightarrow 2$, $\ln|x| \rightarrow 1/x$.
✓

EXAM TECHNIQUE

Don't try to integrate $(\sqrt{x} + 1/\sqrt{x})^2$ in one go. Expand it first: the middle term is just 2. And "find **all** functions" means the + C carries marks.

Marks	Awarded for
3	Expanding $(\sqrt{x} + 1/\sqrt{x})^2 = x + 2 + 1/x$.
2.5	$4 \sin(2x) \rightarrow -2 \cos(2x)$.
2.5	$-e^{2x} \rightarrow -\frac{1}{2}e^{2x}$.
3.5	$x + 2 + 1/x \rightarrow x^2/2 + 2x + \ln x $.
1.5	+ C included (all functions).

1(b)**[12 marks]**Find the function $g(x)$ if $g''(x) = 6x^2 - 4x + 5$, $g(0) = 0$, and $g'(0) = 1$.**ANSWER**

$$g(x) = \frac{1}{2}x^4 - \frac{2}{3}x^3 + \frac{5}{2}x^2 + x$$

RULES YOU NEED

- $\int x^n dx = x^{n+1}/(n+1)$: add 1 to the power, divide by the new power.
- Integrate g'' to get g' , then integrate g' to get g . Each integration gives a new constant.
- Use the condition that matches each step: $g'(0)$ finds the first constant, $g(0)$ the second.

WORKING — STEP BY STEP**1 Integrate $g''(x)$ to get $g'(x)$**

Term by term:

$$6x^2 \rightarrow 6 \cdot \frac{x^3}{3} = 2x^3 \quad -4x \rightarrow -4 \cdot \frac{x^2}{2} = -2x^2 \quad 5 \rightarrow 5x$$

$$g'(x) = 2x^3 - 2x^2 + 5x + C$$

2 Use $g'(0) = 1$ to find C Put $x = 0$ into $g'(x)$:

$$g'(0) = 2(0)^3 - 2(0)^2 + 5(0) + C = C$$

$$C = 1 \Rightarrow g'(x) = 2x^3 - 2x^2 + 5x + 1$$

3 Integrate $g'(x)$ to get $g(x)$ Term by term again, with a new constant D :

$$2x^3 \rightarrow 2 \cdot \frac{x^4}{4} = \frac{1}{2}x^4 \quad -2x^2 \rightarrow -2 \cdot \frac{x^3}{3} = -\frac{2}{3}x^3$$

$$5x \rightarrow 5 \cdot \frac{x^2}{2} = \frac{5}{2}x^2 \quad 1 \rightarrow x$$

$$g(x) = \frac{1}{2}x^4 - \frac{2}{3}x^3 + \frac{5}{2}x^2 + x + D$$

4 Use $g(0) = 0$ to find D

$$g(0) = 0 - 0 + 0 + 0 + D = D \Rightarrow D = 0$$

5 Check (optional)

Differentiate twice: $g' = 2x^3 - 2x^2 + 5x + 1$ (so $g'(0) = 1$ ✓), $g'' = 6x^2 - 4x + 5$ ✓.

EXAM TECHNIQUE

Find C straight after the first integration, before integrating again. If you carry C along unfound, it turns into Cx and is easy to lose.

Marks	Awarded for
3	$g'(x) = 2x^3 - 2x^2 + 5x + C$.
2.5	$C = 1$ from $g'(0) = 1$.
3.5	Integrating again: $\frac{1}{2}x^4 - \frac{2}{3}x^3 + (5/2)x^2 + x + D$.
2	$D = 0$ from $g(0) = 0$.
1	Final $g(x)$ stated.

Question 1 Total: 25 Marks

Question 2

Integration by substitution, integration by parts — 25 marks

2(a)

[11 marks]

Use integration by substitution to evaluate $\int \sqrt{x} \sin(1 + x^{3/2}) dx$.

ANSWER

$$-\frac{2}{3} \cos(1 + x^{3/2}) + C$$

RULES YOU NEED

- Substitution: let $u =$ the “inside” function, then $du = (du/dx) dx$.
- Power rule: $d/dx (x^{3/2}) = (3/2)x^{1/2} = (3/2)\sqrt{x}$.
- $\int \sin u du = -\cos u$.

WORKING — STEP BY STEP

1 Choose u

Pick the inside of the sin. Its derivative has a $\sqrt{x}$ in it, and $\sqrt{x}$ is sitting outside, which is the sign this will work:

$$u = 1 + x^{3/2}$$

2 Find du

Differentiate: the 1 goes to 0, and bring the power $3/2$ down, taking 1 off ($3/2 - 1 = 1/2$):

$$\frac{du}{dx} = \frac{3}{2}x^{1/2} = \frac{3}{2}\sqrt{x} \Rightarrow du = \frac{3}{2}\sqrt{x} dx$$

3 Get $\sqrt{x} dx$ on its own

Multiply both sides by $2/3$:

$$\sqrt{x} dx = \frac{2}{3} du$$

4 Substitute

Replace $1 + x^{3/2}$ by u and $\sqrt{x} dx$ by $(2/3) du$:

$$\int \sin(u) \cdot \frac{2}{3} du = \frac{2}{3} \int \sin u du$$

5 Integrate

$$\frac{2}{3} (-\cos u) + C = -\frac{2}{3} \cos u + C$$

6 Put x back in

Replace u with $1 + x^{3/2}$:

$$-\frac{2}{3} \cos(1 + x^{3/2}) + C$$

EXAM TECHNIQUE

Always finish in terms of x : an answer left as $-(2/3) \cos u$ loses marks. Check by differentiating: the chain rule gives $-(2/3) \cdot (-\sin(...)) \cdot (3/2)\sqrt{x} = \sqrt{x} \sin(1 + x^{3/2}) \checkmark$.

Marks	Awarded for
3	$u = 1 + x^{3/2}$ and $du = (3/2)\sqrt{x} dx$.
3	$\sqrt{x} dx = (2/3) du$, integral rewritten in u .
3	$\int \sin u du = -\cos u$, giving $-(2/3) \cos u$.
2	Back in terms of x , with $+ C$.

2(b)**[14 marks]**

Use integration by parts to evaluate $\int x^2 \sin(\pi x) dx$.

ANSWER

$$-\frac{x^2}{\pi} \cos(\pi x) + \frac{2x}{\pi^2} \sin(\pi x) + \frac{2}{\pi^3} \cos(\pi x) + C$$

RULES YOU NEED

- Integration by parts: $\int u v' dx = u v - \int u' v dx$.
- Let u = the power of x : it gets simpler each time you differentiate it.
- $\int \sin(ax) dx = -(1/a) \cos(ax)$ and $\int \cos(ax) dx = (1/a) \sin(ax)$. Here $a = \pi$.

WORKING — STEP BY STEP**1 Choose u and v'**

Let $u = x^2$ (differentiating makes it simpler) and $v' = \sin(\pi x)$:

$$u = x^2 \Rightarrow u' = 2x \quad v' = \sin(\pi x) \Rightarrow v = -\frac{1}{\pi} \cos(\pi x)$$

2 Apply the parts formula (first time)

$uv - \int u'v dx$. The two minus signs in the last bit make a plus:

$$\int x^2 \sin(\pi x) dx = x^2 \cdot \left(-\frac{1}{\pi} \cos(\pi x)\right) - \int 2x \cdot \left(-\frac{1}{\pi} \cos(\pi x)\right) dx$$

$$= -\frac{x^2}{\pi} \cos(\pi x) + \frac{2}{\pi} \int x \cos(\pi x) dx$$

3 Parts again on $\int x \cos(\pi x) dx$

There is still an x inside, so use parts a second time with $u = x$:

$$u = x \Rightarrow u' = 1 \quad v' = \cos(\pi x) \Rightarrow v = \frac{1}{\pi} \sin(\pi x)$$

$$\int x \cos(\pi x) dx = \frac{x}{\pi} \sin(\pi x) - \int \frac{1}{\pi} \sin(\pi x) dx$$

4 Finish the second integral

$\int \sin(\pi x) dx = -(1/\pi) \cos(\pi x)$, so:

$$\int \frac{1}{\pi} \sin(\pi x) dx = \frac{1}{\pi} \cdot \left(-\frac{1}{\pi} \cos(\pi x)\right) = -\frac{1}{\pi^2} \cos(\pi x)$$

$$\int x \cos(\pi x) dx = \frac{x}{\pi} \sin(\pi x) + \frac{1}{\pi^2} \cos(\pi x)$$

5 Put it back into step 2

Multiply each term of step 4 by $2/\pi$:

$$\frac{2}{\pi} \cdot \frac{x}{\pi} \sin(\pi x) = \frac{2x}{\pi^2} \sin(\pi x) \quad \frac{2}{\pi} \cdot \frac{1}{\pi^2} \cos(\pi x) = \frac{2}{\pi^3} \cos(\pi x)$$

$$\int x^2 \sin(\pi x) dx = -\frac{x^2}{\pi} \cos(\pi x) + \frac{2x}{\pi^2} \sin(\pi x) + \frac{2}{\pi^3} \cos(\pi x) + C$$

EXAM TECHNIQUE

Use brackets at every stage. Nearly all lost marks here are sign slips: $-\int 2x \cdot (-(1/\pi) \cos)$ becomes $+(2/\pi) \int x \cos$.

Marks	Awarded for
3	$u = x^2$, $v = -(1/\pi) \cos(\pi x)$ (correct choice and v).
3	First application: $-(x^2/\pi) \cos(\pi x) + (2/\pi) \int x \cos(\pi x) dx$.
4	Second application with $u = x$, $v = (1/\pi) \sin(\pi x)$.
2	$\int \sin(\pi x) dx = -(1/\pi) \cos(\pi x)$ handled correctly.
2	Final answer combined correctly, $+ C$.

Question 2 Total: 25 Marks

Question 3

Definite integrals (by parts), area between two curves — 25 marks

3(a)

[13 marks]

Evaluate $\int_1^2 \frac{\ln x}{x^2} dx$.

ANSWER

$$\frac{1 - \ln 2}{2} \approx 0.153$$

RULES YOU NEED

- Parts with limits: $\int_a^b u v' dx = [u v]_a^b - \int_a^b u' v dx$.
- $d/dx (\ln x) = 1/x$, and $\int x^{-2} dx = x^{-1}/(-1) = -1/x$.
- $\ln 1 = 0$.

WORKING — STEP BY STEP

1 Choose u and v'

We can't integrate $\ln x$ directly, but we can differentiate it, so $u = \ln x$. Write $1/x^2$ as x^{-2} :

$$u = \ln x \Rightarrow u' = \frac{1}{x} \quad v' = x^{-2} \Rightarrow v = \frac{x^{-1}}{-1} = -\frac{1}{x}$$

2 Apply the parts formula

$u'v = (1/x)(-1/x) = -1/x^2$, and minus a minus is a plus:

$$\int_1^2 \frac{\ln x}{x^2} dx = \left[-\frac{\ln x}{x} \right]_1^2 - \int_1^2 \frac{1}{x} \cdot \left(-\frac{1}{x} \right) dx$$

$$= \left[-\frac{\ln x}{x} \right]_1^2 + \int_1^2 \frac{1}{x^2} dx$$

3 Integrate the last part

$\int 1/x^2 dx = -1/x$, so both pieces go in one bracket:

$$= \left[-\frac{\ln x}{x} - \frac{1}{x} \right]_1^2$$

4 Put in the limits

Top limit $x = 2$, then bottom limit $x = 1$ (remember $\ln 1 = 0$):

$$x = 2: -\frac{\ln 2}{2} - \frac{1}{2}$$

$$x = 1: -\frac{\ln 1}{1} - \frac{1}{1} = -0 - 1 = -1$$

5 Subtract: top – bottom

$$\left(-\frac{\ln 2}{2} - \frac{1}{2}\right) - (-1) = -\frac{\ln 2}{2} - \frac{1}{2} + 1 = \frac{1}{2} - \frac{\ln 2}{2} = \frac{1 - \ln 2}{2}$$

6 Decimal (optional)

$$\ln 2 \approx 0.6931 \Rightarrow \frac{1 - 0.6931}{2} = \frac{0.3069}{2} \approx 0.153$$

EXAM TECHNIQUE

Subtracting the bottom limit: $(...) - (-1)$ is **+ 1**. Also, the answer should be positive: $\ln x \geq 0$ on $[1, 2]$, so the area under the curve can't be negative.

Marks	Awarded for
3	$u = \ln x$, $v = -1/x$.
3	Parts formula applied correctly with limits.
2	$\int 1/x^2 \, dx = -1/x$.
3	Limits substituted correctly (using $\ln 1 = 0$).
2	Final answer $(1 - \ln 2)/2$.

3(b)**[12 marks]**

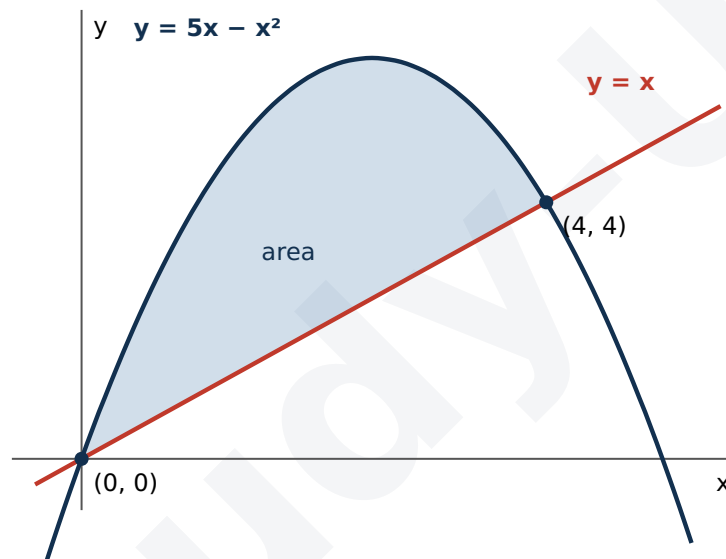
Compute the area of the region between the curves $y = 5x - x^2$ and $y = x$.

ANSWER

$$\text{Area} = \frac{32}{3} \approx 10.67 \text{ square units}$$

RULES YOU NEED

- Area between two curves = $\int (\text{upper curve} - \text{lower curve}) dx$.
- The limits are the x-values where the curves meet.



The parabola is on top between $x = 0$ and $x = 4$. The shaded region is the area we want.

WORKING — STEP BY STEP**1 Find where the curves meet**

Set the two equal, bring everything to one side and factor:

$$5x - x^2 = x \Rightarrow 4x - x^2 = 0 \Rightarrow x(4 - x) = 0$$

$$x = 0 \quad \text{or} \quad x = 4$$

2 Which curve is on top?

Test any x between 0 and 4, e.g. $x = 1$:

$$\text{Parabola: } 5(1) - 1^2 = 4 \quad \text{Line: } y = 1 \Rightarrow \text{parabola on top}$$

3 Set up the integral (upper – lower)

$$A = \int_0^4 [(5x - x^2) - x] dx = \int_0^4 (4x - x^2) dx$$

4 Integrate

$4x \rightarrow 4 \cdot x^2/2 = 2x^2$, and $x^2 \rightarrow x^3/3$:

$$A = \left[2x^2 - \frac{x^3}{3} \right]_0^4$$

5 Put in the limits

$$x = 4: 2(16) - \frac{64}{3} = 32 - \frac{64}{3} = \frac{96}{3} - \frac{64}{3} = \frac{32}{3}$$

$$x = 0: 0$$

$$A = \frac{32}{3} - 0 = \frac{32}{3} \approx 10.67$$

EXAM TECHNIQUE

Always do upper – lower. If you get a negative area, you had them the wrong way round. A quick sketch settles it.

Marks	Awarded for
3	Intersections $x = 0$ and $x = 4$.
2	Identifying $5x - x^2$ as the upper curve.
2	Integral set up: $\int_0^4 (4x - x^2) dx$.
3	Integration: $2x^2 - x^3/3$.
2	Area = $32/3$.

Question 3 Total: 25 Marks

Question 4

Volumes of rotation, trigonometric integrals by substitution — 25 marks

4(a)

[12 marks]

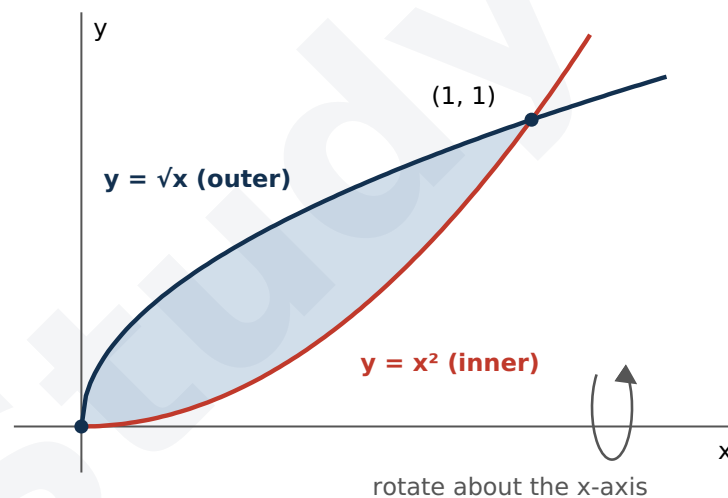
Find the volume of the solid obtained by rotating the region bounded by the curves $y = x^2$ and $y^2 = x$ about the x-axis.

ANSWER

$$V = \frac{3\pi}{10} \text{ cubic units } (\approx 0.942)$$

RULES YOU NEED

- Volume of revolution about the x-axis: $V = \int_a^b \pi y^2 dx$.
- A region between two curves makes a solid with a hole: $V = (\text{volume from the outer curve}) - (\text{volume from the inner curve})$.



The shaded region between $y = \sqrt{x}$ and $y = x^2$ spins around the x-axis. It makes a solid with a hollow core.

WORKING — STEP BY STEP

1 Write both curves as $y = \dots$

$y^2 = x$ gives $y = \sqrt{x}$ (the top half, which is the part that bounds the region):

$$y = \sqrt{x} \quad \text{and} \quad y = x^2$$

2 Find where they meet

Set them equal and square both sides:

$$x^2 = \sqrt{x} \Rightarrow x^4 = x \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0$$

$$x = 0 \quad \text{or} \quad x = 1$$

3 Which is further from the axis?

Test $x = \frac{1}{2}$:

$$\sqrt{0.5} \approx 0.71 \quad (0.5)^2 = 0.25 \Rightarrow \text{outer: } y = \sqrt{x}, \text{ inner: } y = x^2$$

4 Set up the volume

Use $V = \int \pi y^2 dx$ for each curve and subtract the hole. Square each curve on its own:

$$V = \int_0^1 \pi (\sqrt{x})^2 dx - \int_0^1 \pi (x^2)^2 dx = \pi \int_0^1 (x - x^4) dx$$

5 Integrate

$$\pi \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1$$

6 Put in the limits

$$\pi \left[\left(\frac{1}{2} - \frac{1}{5} \right) - 0 \right] = \pi \left(\frac{5}{10} - \frac{2}{10} \right) = \frac{3\pi}{10}$$

EXAM TECHNIQUE

Square each curve separately: $(\sqrt{x})^2 - (x^2)^2 = x - x^4$. Writing $(\sqrt{x} - x^2)^2$ is a very common mistake and gives the wrong volume.

Marks	Awarded for
2	Intersections $x = 0$ and $x = 1$.
2	Outer $y = \sqrt{x}$, inner $y = x^2$ identified.
3	$V = \pi \int_0^1 (x - x^4) dx$ set up (outer ² – inner ²).
3	Integration: $x^2/2 - x^5/5$.
2	$V = 3\pi/10$.

4(b)**[13 marks]**Evaluate $\int_{\pi/2}^{3\pi/4} \sin^5 x \cos^3 x \, dx$.**ANSWER**

$$-\frac{11}{384} \approx -0.0286$$

RULES YOU NEED

- $\cos^2 x = 1 - \sin^2 x$ (from $\cos^2 x + \sin^2 x = 1$).
- Substitution $u = \sin x \Rightarrow du = \cos x \, dx$. Change the limits to u -values too.
- $\sin(\pi/2) = 1$ and $\sin(3\pi/4) = 1/\sqrt{2}$.

WORKING — STEP BY STEP**1 Split off one $\cos x$**

$\cos$ has an odd power (3). Keep one $\cos x$ for du and change the other two into $\sin$:

$$\cos^3 x = \cos^2 x \cdot \cos x = (1 - \sin^2 x) \cos x$$

2 Rewrite the integral

$$\int_{\pi/2}^{3\pi/4} \sin^5 x (1 - \sin^2 x) \cos x \, dx$$

3 Substitute $u = \sin x$

$du = \cos x \, dx$, so the $\cos x \, dx$ becomes du :

$$\int u^5 (1 - u^2) \, du = \int (u^5 - u^7) \, du$$

4 Change the limits

$3\pi/4$ is 135° , and $\sin 135^\circ = \sin 45^\circ = 1/\sqrt{2}$:

$$x = \pi/2 \Rightarrow u = \sin(\pi/2) = 1 \quad x = 3\pi/4 \Rightarrow u = \sin(3\pi/4) = 1/\sqrt{2}$$

$$\int_1^{1/\sqrt{2}} (u^5 - u^7) \, du = \left[\frac{u^6}{6} - \frac{u^8}{8} \right]_1^{1/\sqrt{2}}$$

5 Work out the powers of $1/\sqrt{2}$

$(1/\sqrt{2})^2 = 1/2$, so:

$$(1/\sqrt{2})^6 = \left(\frac{1}{2}\right)^3 = \frac{1}{8} \quad (1/\sqrt{2})^8 = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

6 Put in the limits

Common denominator 384 ($= 48 \times 8 = 128 \times 3$):

$$u = 1/\sqrt{2}: \quad \frac{1/8}{6} - \frac{1/16}{8} = \frac{1}{48} - \frac{1}{128} = \frac{8}{384} - \frac{3}{384} = \frac{5}{384}$$

$$u = 1: \quad \frac{1}{6} - \frac{1}{8} = \frac{4}{24} - \frac{3}{24} = \frac{1}{24} = \frac{16}{384}$$

7 Subtract: top – bottom

$$\frac{5}{384} - \frac{16}{384} = -\frac{11}{384} \approx -0.0286$$

A negative answer makes sense: between $\pi/2$ and $3\pi/4$, $\cos x$ is negative, so $\cos^3 x < 0$ while $\sin^5 x > 0$.

EXAM TECHNIQUE

Keep the limits in the order they come from x : the top limit is $1/\sqrt{2}$ even though it is smaller than 1. Swapping them flips the sign.

Marks	Awarded for
3	$\cos^3 x = (1 - \sin^2 x) \cos x$.
2	$u = \sin x$, $du = \cos x \, dx$; integral becomes $\int (u^5 - u^7) \, du$.
2	New limits $u = 1$ and $u = 1/\sqrt{2}$.
2	Integration: $u^6/6 - u^8/8$.
3	Limits evaluated: $5/384$ and $1/24$.
1	Final answer $-11/384$.

Question 4 Total: 25 Marks

Question 5

Separable differential equations (partial fractions), line of best fit — 25 marks

5(a)

[15 marks]

Solve the differential equation $x \frac{dy}{dx} + y = y^2$, given the condition $y(1) = -1$.

ANSWER

$$y = \frac{1}{1 - 2x}$$

RULES YOU NEED

- Separable equation: get every y (with dy) on one side and every x (with dx) on the other, then integrate both sides.
- Partial fractions: $1/[y(y - 1)] = A/y + B/(y - 1)$.
- $\int 1/x \, dx = \ln|x|$, and $\ln a - \ln b = \ln(a/b)$.

WORKING — STEP BY STEP

1 Rearrange

Take y from both sides and factor:

$$x \frac{dy}{dx} = y^2 - y = y(y - 1)$$

2 Separate the variables

Divide by $y(y - 1)$ and by x , then multiply by dx :

$$\frac{1}{y(y - 1)} \, dy = \frac{1}{x} \, dx$$

3 Partial fractions

Write the left side as two simple fractions. Multiply through by $y(y - 1)$:

$$\frac{1}{y(y - 1)} = \frac{A}{y} + \frac{B}{y - 1} \Rightarrow 1 = A(y - 1) + By$$

$$y = 0: 1 = A(-1) \Rightarrow A = -1 \quad y = 1: 1 = B(1) \Rightarrow B = 1$$

$$\frac{1}{y(y - 1)} = \frac{1}{y - 1} - \frac{1}{y}$$

4 Integrate both sides

$$\int \left(\frac{1}{y-1} - \frac{1}{y} \right) dy = \int \frac{1}{x} dx$$

$$\ln|y-1| - \ln|y| = \ln|x| + C$$

5 Remove the logs

Combine the logs, then take e of both sides. The constant e^C (with the $\pm$ from the absolute values) becomes one new constant K:

$$\ln\left|\frac{y-1}{y}\right| = \ln|x| + C \Rightarrow \frac{y-1}{y} = Kx$$

6 Use $y(1) = -1$ to find K

Put $x = 1$ and $y = -1$:

$$\frac{-1-1}{-1} = K(1) \Rightarrow \frac{-2}{-1} = K \Rightarrow K = 2$$

$$\frac{y-1}{y} = 2x$$

7 Solve for y

Multiply by y, collect the y terms, then factor out y:

$$y-1 = 2xy \Rightarrow y-2xy = 1 \Rightarrow y(1-2x) = 1$$

$$y = \frac{1}{1-2x}$$

8 Check

$$y(1) = \frac{1}{1-2} = -1 \checkmark$$

EXAM TECHNIQUE

Find K while the equation is still in the form $(y-1)/y = Kx$. It is much quicker than solving for y first.

Marks	Awarded for
3	Separated: $dy/[y(y-1)] = dx/x$.
4	Partial fractions $A = -1$, $B = 1$.
3	Integrated: $\ln y-1 - \ln y = \ln x + C$.
3	Logs removed and $K = 2$ from $y(1) = -1$.
2	$y = 1/(1-2x)$.

5(b)**[10 marks]**

Recall, for a collection of n points given by (x_i, y_i) , the line of best fit is a line of the form $y = mx + c$, with $m = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$, $c = \frac{\sum y_i - m\sum x_i}{n}$. Use these formulas to compute the line of best fit for the points below, giving m and c correct to 1 decimal place.

x : 2, 3, 5, 7, 9 y : 4, 5, 7, 10, 15

ANSWER

$$m \approx 1.5, \quad c \approx 0.3 \quad \Rightarrow \quad y = 1.5x + 0.3$$

NOTE

Line of best fit isn't one of the listed MA1002 course topics. The question gives you both formulas, so it only needs careful arithmetic.

RULES YOU NEED

- Make a table with columns for x , y , xy and x^2 , then add up each row.
- $\sum x^2$ means square each x , then add. $(\sum x)^2$ means add the x 's, then square. They are different.
- Use the unrounded m when working out c . Round only at the very end.

WORKING — STEP BY STEP**1 Build the table of sums**

Here $n = 5$ (five points). Multiply each x by its y , and square each x :

	1	2	3	4	5	Sum (Σ)
x	2	3	5	7	9	26
y	4	5	7	10	15	41
xy	8	15	35	70	135	263
x^2	4	9	25	49	81	168

2 Top of the m formula

$$n\sum xy - \sum x \sum y = 5 \times 263 - 26 \times 41 = 1315 - 1066 = 249$$

3 Bottom of the m formula

$$n\sum x^2 - (\sum x)^2 = 5 \times 168 - 26^2 = 840 - 676 = 164$$

4 Find m

$$m = \frac{249}{164} = 1.5183... \approx 1.5$$

5 Find c (using the full m)

$$m\Sigma x = \frac{249}{164} \times 26 = \frac{6474}{164} = 39.476...$$

$$c = \frac{41 - 39.476...}{5} = \frac{1.524...}{5} = 0.3049... \approx 0.3$$

6 Write the line

$$y = 1.5x + 0.3$$

EXAM TECHNIQUE

If you round m to 1.5 **before** finding c, you get $c = (41 - 39)/5 = 0.4$, which is wrong to 1 decimal place. Keep m unrounded until the end.

Marks	Awarded for
4	Sums $\Sigma x = 26$, $\Sigma y = 41$, $\Sigma xy = 263$, $\Sigma x^2 = 168$ (1 each).
3	$m = 249/164 \approx 1.5$.
2	$c \approx 0.3$ using the unrounded m.
1	Line $y = 1.5x + 0.3$ stated.

Question 5 Total: 25 Marks